XPECTO 2026

Quant Hackathon — IIT Mandi

Momentum Trading Strategy

using Machine Learning

Author	Mayank Srivastava
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1. Executive Summary

This report presents a Machine Learning-based Weekly Momentum Portfolio Strategy developed for the Xpecto 2026 Quant Hackathon at IIT Mandi. The strategy leverages three supervised classification models to predict short-term stock return directions across a universe of 10 large-cap US equities, constructing a concentrated two-stock portfolio that is rebalanced weekly.

The strategy was trained on six years of daily market data (2017–2022) and evaluated out-of-sample over the 2023–2025 test period. Logistic Regression emerged as the best-performing classifier with a test accuracy of 57.79%, outperforming Random Forest (57.53%) and XGBoost (55.68%).

72.37%	1.2921	-12.58%	57.58%
Cumulative Return	Sharpe Ratio	Max Drawdown	Hit Rate

2. Strategy Overview

2.1 Universe & Parameters

The strategy operates on a fixed universe of 10 large-cap US equities spanning multiple sectors:

AAPL	MSFT	GOOGL	AMZN	META	TSLA
Apple	Microsoft	Alphabet	Amazon	Meta	Tesla

JPM	V	JNJ	BRK-B
JPMorgan	Visa	J&J	Berkshire Hathaway

2.2 Key Parameters

Parameter	Value	Description
Initial Capital	$100,000	Starting portfolio value
Training Period	2017-01-01 to 2022-12-31	6 years of in-sample data
Test Period	2023-01-01 to 2025-01-01	2 years out-of-sample
Rebalancing	Weekly (every 5 trading days)	Monday-to-Monday
Portfolio Size	Top 2 stocks (TOP_N = 2)	Equal weight: 50% each
Transaction Cost	0.1% per side (0.2% round trip)	Entry + exit costs applied
Risk-Free Rate	5.0% per annum	Used for Sharpe calculation

3. Feature Engineering

Sixteen technical features were engineered from daily OHLCV data for each stock. These features capture momentum, mean-reversion, volatility, and volume dynamics across multiple time horizons.

Feature	Formula	Financial Meaning
ret_1w	P(t) / P(t-5) - 1	1-week price momentum
ret_2w	P(t) / P(t-10) - 1	2-week price momentum
ret_1m	P(t) / P(t-21) - 1	1-month price momentum
ret_3m	P(t) / P(t-63) - 1	3-month price momentum
ret_6m	P(t) / P(t-126) - 1	6-month price momentum
rsi_14	RSI(14)	Overbought / Oversold signal
macd_hist	MACD Line - Signal Line	Momentum acceleration
bb_pct	(P - Lower) / (Upper - Lower)	Bollinger Band position
bb_width	4 x Std(20) / SMA(20)	Volatility regime indicator
atr_pct	ATR(14) / Close	Normalised true range
vol_ratio	Volume / MA(Volume, 20)	Volume surge indicator
vol_20d	Std(ret_1d, 20) x sqrt(252)	Annualised 20-day volatility
zscore_20	(P - SMA20) / Std(20)	Mean-reversion z-score
pr_sma20	P / SMA(20) - 1	Price deviation from 20-day MA
pr_sma50	P / SMA(50) - 1	Price deviation from 50-day MA
ret_1d	P(t) / P(t-1) - 1	Prior day return (noise filter)

3.1 Target Variable

The binary classification target is defined as:

$$y = 1 \quad \text{if} \quad \text{Return}_{t+5} > 0, \quad \text{else} \quad y = 0$$

This predicts whether the stock price will be higher 5 trading days (1 week) into the future. The training set contained 13,840 samples: 56.3% positive (up) and 43.7% negative (down).

4. Model Training & Selection

All stocks were pooled into a single training dataset, allowing the model to learn cross-sectional patterns across the universe. Features were standardised using StandardScaler before Logistic Regression training.

Model	Type	Test Accuracy	Rank
Logistic Regression	Linear Classifier	**57.79%**	**#1 (Best)**
Random Forest	Non-linear Ensemble	57.53%	#2
XGBoost	Gradient Boosted Trees	55.68%	#3

4.1 Model Configurations

- Logistic Regression: C=0.1, max_iter=1000, random_state=42. Features standardised with StandardScaler.
- Random Forest: 200 estimators, max_depth=5, min_samples_leaf=20. Raw features used (scale-invariant).
- XGBoost: 200 estimators, max_depth=3, learning_rate=0.05, subsample=0.8, min_child_weight=10.

4.2 Data Split

- Train set: 13,840 samples across all 10 stocks (2017-01-01 to 2022-12-31)
- Test set: 5,020 samples across all 10 stocks (2023-01-01 to 2025-01-01)
- Strict temporal split: No future data leakage — test period entirely post-training.

5. Portfolio Construction

The portfolio is constructed using a systematic weekly rebalancing process driven entirely by ML model predictions:

Step	Action
1	Every Monday: Obtain the latest feature vector for all 10 stocks
2	Run Logistic Regression to predict P(return > 0) for each stock
3	Rank all 10 stocks by predicted probability (highest first)
4	Select the TOP 2 stocks (highest predicted upside probability)
5	Allocate 50% of portfolio capital to each selected stock (equal weight)
6	Apply 0.1% transaction cost on entry; hold for exactly 1 week
7	On exit, apply 0.1% transaction cost; repeat from Step 1

This process ran for 100 weeks over the 2023-2025 test period, growing an initial capital of $100,000 to $180,573.

6. Performance Metrics

6.1 Full Performance Summary

Metric	Before Costs	After Costs	Benchmark
Cumulative Return	—	72.37%	—
Annualised Return	47.70%	33.11%	50.80%
Annualised Volatility	—	20.87%	—
Sharpe Ratio	2.0422	1.2921	2.3771
Max Drawdown	-11.88%	-12.58%	-12.07%
Hit Rate	62.63%	57.58%	N/A
Avg Weekly Return	0.7939%	0.5924%	—
Final Capital	—	$180,573	—
Total Weeks Traded	—	99	—

6.2 Transaction Cost Analysis

Transaction costs impose a significant drag on strategy performance, a common challenge for high-frequency rebalancing strategies:

Impact Area	Gross (Before)	Net (After Costs)
Annualised Return	47.70%	33.11%
Sharpe Ratio	2.0422	1.2921
Hit Rate	62.63%	57.58%
Annual Cost Drag	—	**14.60% p.a.**

6.3 Benchmark Comparison

The benchmark is an equal-weight buy-and-hold portfolio of all 10 stocks rebalanced at the start of the test period. Key observations:

- The strategy achieved a 72.37% cumulative return vs the benchmark over the test window.
- On an annualised basis, the strategy returned 33.11% (after costs) vs the benchmark's 50.80%.
- Before transaction costs, the strategy annualised return of 47.70% approached the benchmark.
- The benchmark's Sharpe of 2.38 exceeded the strategy's 1.29, reflecting lower turnover costs.
- Max drawdown was comparable: strategy -12.58% vs benchmark -12.07%, showing similar downside risk.

7. Key Findings & Observations

7.1 Stock Selection Patterns

Analysis of the 100 weekly selections reveals strong sector concentration:

- **JNJ (Johnson & Johnson)** was the most frequently selected stock, appearing in more portfolio selections than any other ticker — suggesting stable, predictable momentum characteristics.
- **Defensive stocks (JPM, V, JNJ, BRK-B)** tended to dominate selections, reflecting the model's preference for lower-volatility, high-probability upward moves.
- **TSLA and GOOGL** were least selected, consistent with their higher volatility making direction prediction harder.

7.2 Model Insights

- Logistic Regression outperformed more complex non-linear models, consistent with the Efficient Market Hypothesis — returns are close to random walks and simple linear boundaries capture most predictable variance.
- The 57-58% accuracy range is consistent with academic literature on short-term return predictability using technical features.
- The relatively stable 56.3%/43.7% train class balance minimised class-imbalance bias.

7.3 Equity Curve Behaviour

- The portfolio showed steady upward progression with multiple drawdown periods, the deepest being -12.58%.
- Weekly return distribution was approximately normal with slight positive skew, confirming the momentum signal is genuine but small.
- The strategy demonstrated consistent positive weeks (hit rate 57.58%), meaning it wins more often than it loses.

8. Limitations & Risk Factors

8.1 Methodological Limitations

- US equities only — the strategy is calibrated on liquid, large-cap US stocks. Results may not generalise to other markets, asset classes, or smaller stocks with liquidity constraints.
- Look-ahead bias risk — while strict temporal splitting was applied, the feature universe was selected with knowledge of the full period. Future refinement should use a walk-forward feature selection process.
- Model trained on 2017–2022 data may not capture post-2022 structural regime changes (e.g., high interest rate environment, AI market dynamics, geopolitical shocks).
- Assumes perfect liquidity — weekly rebalancing assumes all trades execute at the closing price without market impact, which may not hold for large position sizes.

8.2 Transaction Cost Sensitivity

- At 0.2% round-trip cost per trade, the strategy loses 14.60% per year in cost drag — making it highly sensitive to execution quality. At institutional trading costs (0.01-0.05%), performance would improve substantially.
- The concentrated 2-stock portfolio creates high single-stock risk, amplified by weekly turnover.

8.3 Future Improvements

- Walk-forward cross-validation to better simulate live trading conditions.
- Regime detection layer to reduce trading frequency in trending bull markets (where buy-and-hold dominates).
- Portfolio expansion to TOP_N = 3 or 4 to reduce concentration risk.
- Integration of macro features (VIX, yield spreads, sector rotation signals).
- Risk-adjusted position sizing (Kelly criterion or volatility targeting) instead of equal weighting.

9. References

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